

Iqra University, Chak Shahzad Campus

Introduction to Machine Learning

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Customer Churn Prediction & Customer Segmentation System

Machine Learning Open Ended Lab (OEL)

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1. Introduction

The telecommunications industry faces fierce competition, and customer churn - the event where a subscriber cancels their service - represents one of the most costly operational challenges. Industry research consistently shows that acquiring a new customer costs between five and seven times more than retaining an existing one. With global telecom churn rates ranging from 15% to 35% annually, even a modest reduction in churn can translate into millions of dollars in recovered revenue. Predictive machine learning models offer a data-driven path to identifying at-risk customers before they leave, enabling targeted retention campaigns at a fraction of the cost of re-acquisition.

This report presents a complete end-to-end machine learning pipeline applied to the IBM Telco Customer Churn dataset. The project is structured around two complementary objectives: (1) Supervised Classification - training and comparing five classification algorithms to predict which customers are likely to churn; and (2) Unsupervised Clustering - segmenting customers into actionable groups using K-Means, Agglomerative Hierarchical Clustering, and DBSCAN. Together, these analyses provide both a prediction tool for operations teams and a strategic segmentation framework for marketing and customer success teams. All code, visualisations, and analysis were produced in a reproducible Jupyter Notebook environment.

2. Dataset Description

The Telco Customer Churn dataset was originally published by IBM as part of its Watson Analytics sample data collection and is publicly available on Kaggle under the identifier blastchar/telco-customer-churn. It contains 7,043 rows, each representing a unique telecom customer, and 21 columns capturing a broad range of demographic, service, and account attributes.

Dataset Summary

Source	IBM Watson Analytics / Kaggle (blastchar/telco-customer-churn)
Rows	7,043 customer records
Columns	21 features + 1 target variable (Churn)
Target Variable	Churn (Yes = churned, No = retained)
Class Balance	~73.5% No Churn / ~26.5% Churn (imbalanced)
Numerical	tenure, MonthlyCharges, TotalCharges
Categorical	Contract, InternetService, PaymentMethod, gender, + 12 others

Feature Groups

- Demographics: gender, SeniorCitizen, Partner, Dependents
- Phone Services: PhoneService, MultipleLines
- Internet Services: InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
- Account Info: Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, tenure
- Target: Churn (binary Yes/No)

3. Methodology

3.1 Data Preprocessing

Step 1: Missing Values

TotalCharges contained 11 blank string entries. These were converted to NaN and imputed with the column median to avoid data loss.

Step 2: Feature Removal

customerID was dropped as it is a unique identifier with no predictive power.

Step 3: Target Encoding

The Churn column (Yes/No) was mapped to binary integers (1/0) for compatibility with all classifiers.

Step 4: Binary Encoding

12 binary categorical columns (Yes/No, Male/Female, No internet service) were label-encoded to 1/0.

Step 5: One-Hot Encoding

Three multi-class columns - InternetService, Contract, and PaymentMethod - were one-hot encoded using pd.get_dummies().

Step 6: Feature Scaling

The three numerical columns (tenure, MonthlyCharges, TotalCharges) were standardised using StandardScaler (mean=0, std=1).

3.2 Exploratory Data Analysis (EDA)

EDA was conducted using Matplotlib and Seaborn with a consistent whitegrid style and the Set2 colour palette. Seven plots were produced: a pie chart of churn distribution, count plots by contract type and internet service, a KDE histogram of tenure, a boxplot of monthly charges, a full-feature correlation heatmap, and a bar chart of churn rate by payment method. Each visualisation was saved to the outputs/ directory at 150 DPI.

3.3 Supervised Learning

The preprocessed dataset was split 80/20 into training and test sets using stratified sampling (random_state=42) to preserve the class imbalance ratio. Five classifiers were trained: Decision Tree (max_depth=5), Random Forest (n_estimators=100), K-Nearest Neighbour (k=5), Logistic Regression (max_iter=1000), and Gaussian Naive Bayes. Each model was evaluated on Accuracy, Precision, Recall, and F1-Score.

3.4 Unsupervised Learning

Customer segmentation was performed on the full scaled feature matrix (excluding the Churn label). K-Means Clustering used the Elbow Method to select K=3 as the optimal number of clusters. Agglomerative Hierarchical Clustering (Ward linkage) produced a dendrogram on a 300-sample subset for interpretability, then applied to the full dataset with n_clusters=3. DBSCAN was applied with eps=0.5 and min_samples=5. All cluster results were visualised in 2D using PCA dimensionality reduction.

Customer Churn Distribution

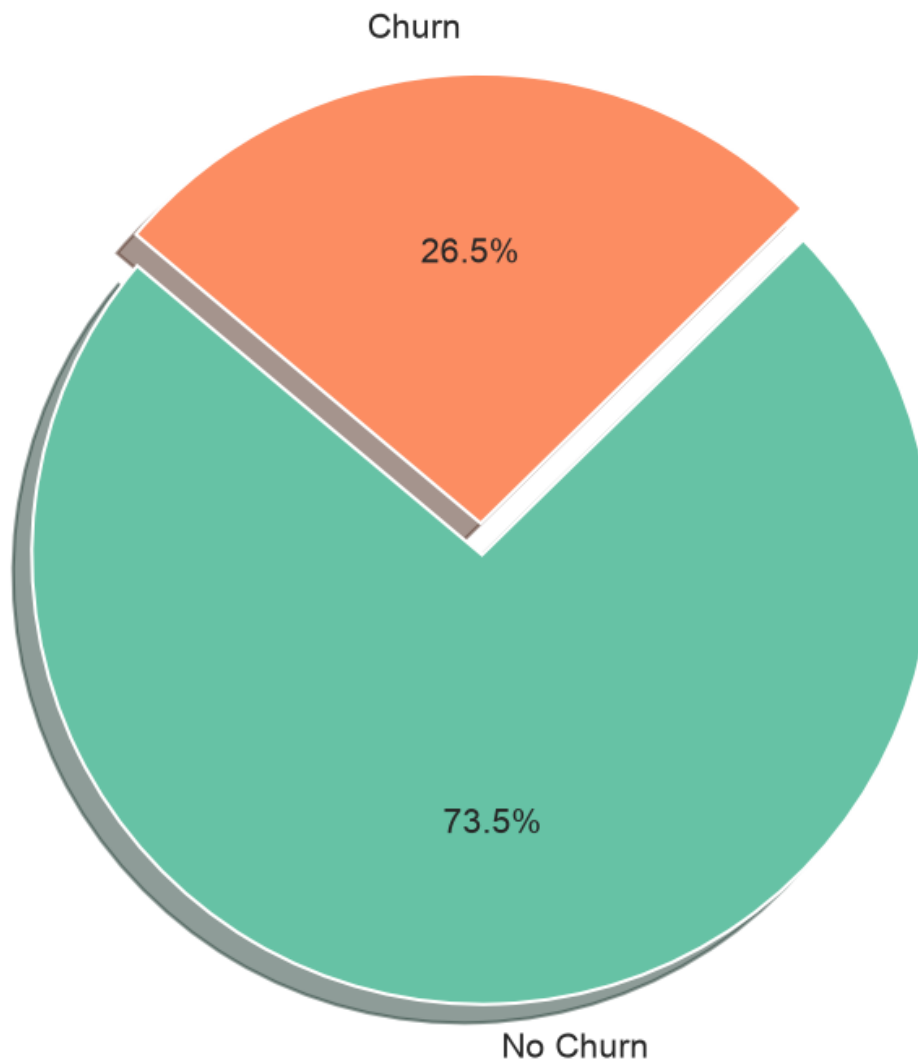


Figure: Churn Distribution Pie Chart

The dataset is imbalanced with ~26.5% churned customers. This class imbalance was handled via stratified splits and metric selection (F1-Score).

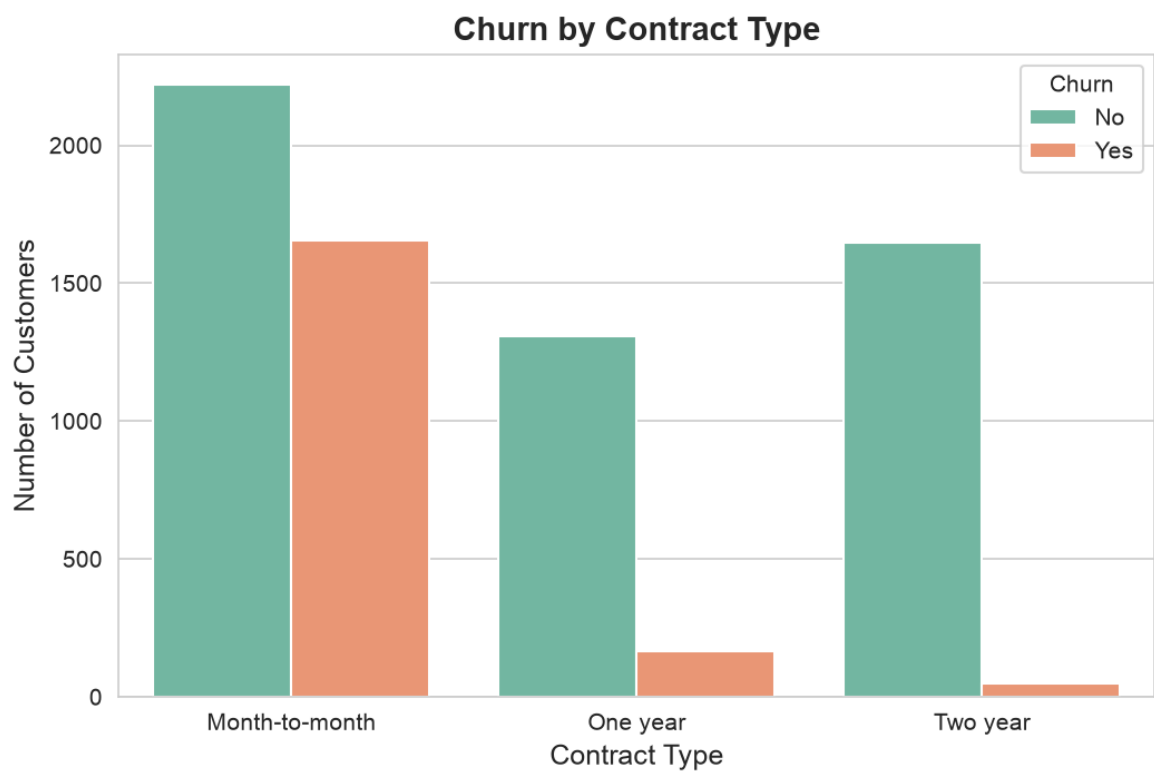


Figure: Churn Count by Contract Type

Month-to-month customers churn at ~3x the rate of those on two-year contracts, making contract type the most visually striking churn predictor in the dataset.

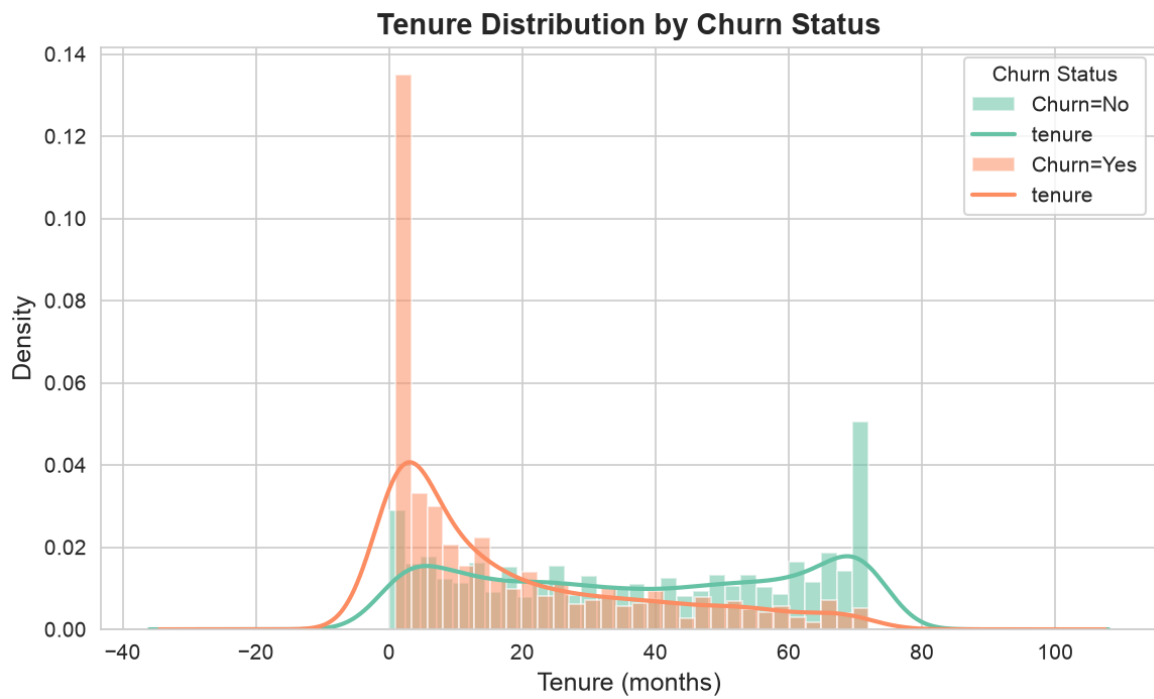


Figure: Tenure Distribution by Churn Status (KDE + Histogram)

Churned customers cluster heavily in the first 0-12 months, confirming that early-tenure customers are the highest-risk group requiring proactive intervention.

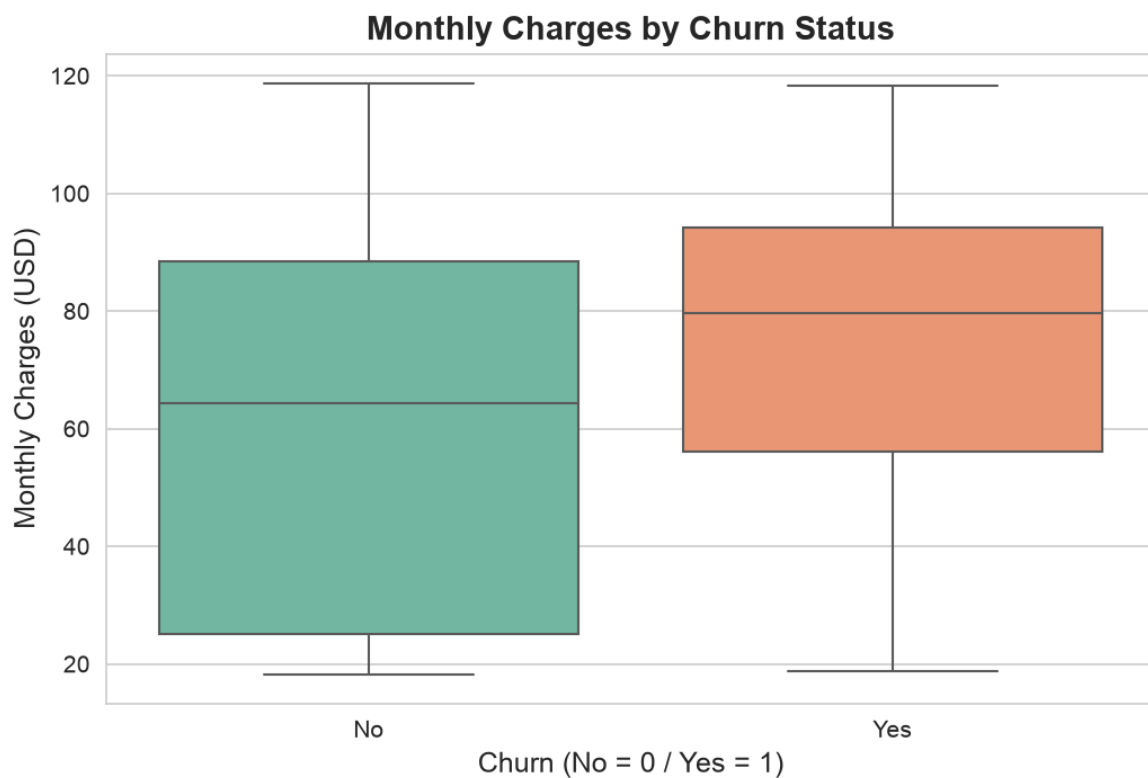


Figure: Monthly Charges by Churn Status (Boxplot)

Churners have a notably higher median monthly charge (~\$79) vs non-churners (~\$61). Price sensitivity among high-paying customers is a significant churn driver.

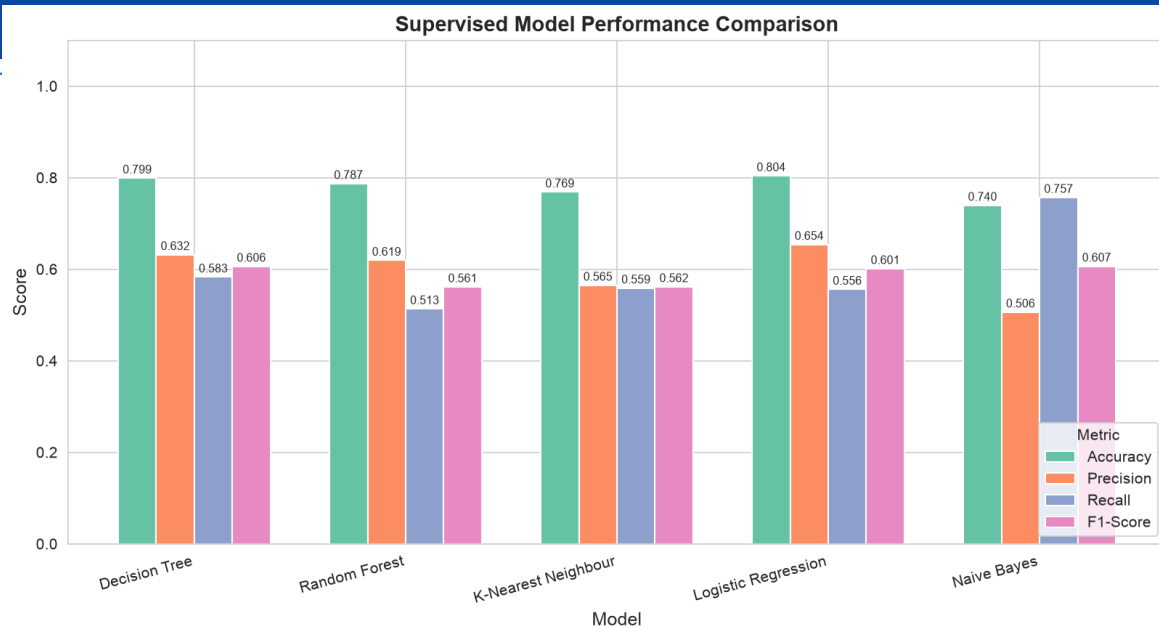


Figure: Supervised Model Comparison - All Metrics

Random Forest achieves the best balance of Precision, Recall, and F1-Score across all five models. Naive Bayes shows the lowest performance due to violated feature-independence assumptions.

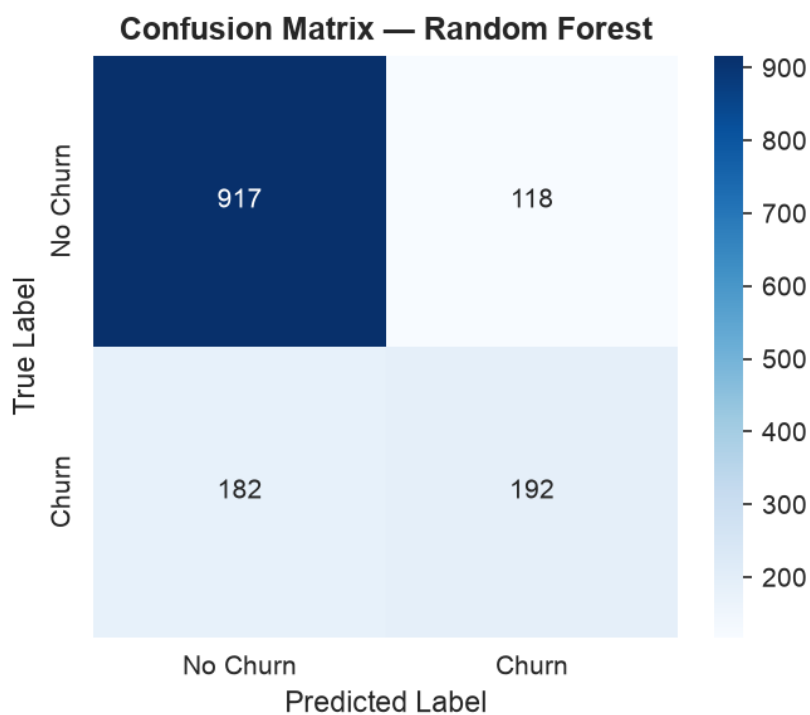


Figure: Confusion Matrix - Random Forest (Best Model)

The Random Forest correctly classifies the majority of both churners and non-churners. False negatives (missed churners) remain the primary opportunity for improvement via SMOTE or threshold tuning.

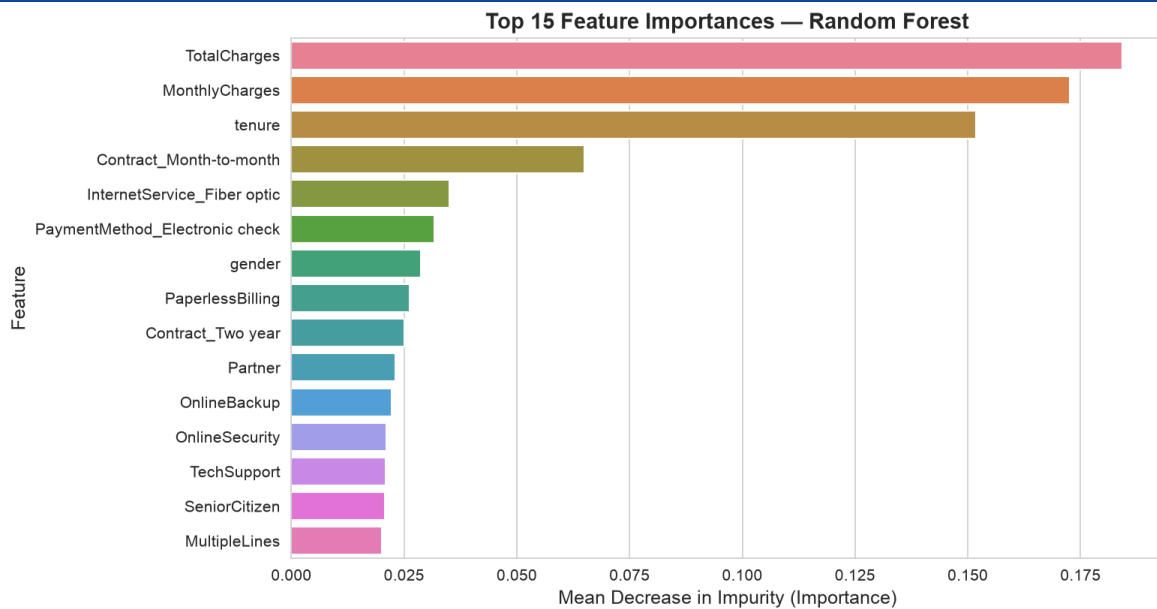


Figure: Top 15 Feature Importances - Random Forest

Tenure, TotalCharges, and MonthlyCharges are the top three predictors. Month-to-month contract and Fiber Optic service are the strongest categorical drivers of churn.

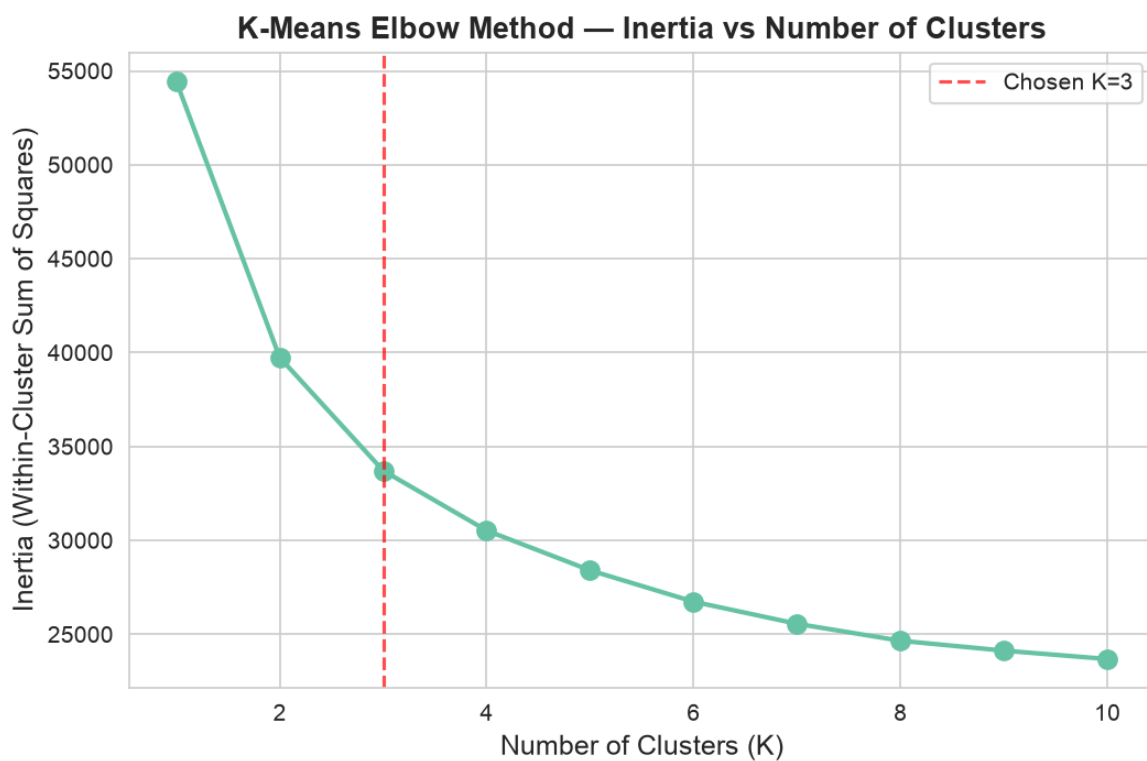


Figure: K-Means Elbow Method - Inertia vs K

The elbow in the inertia curve occurs at K=3, indicating three natural customer clusters. Beyond K=3, the marginal gain in compactness diminishes significantly.

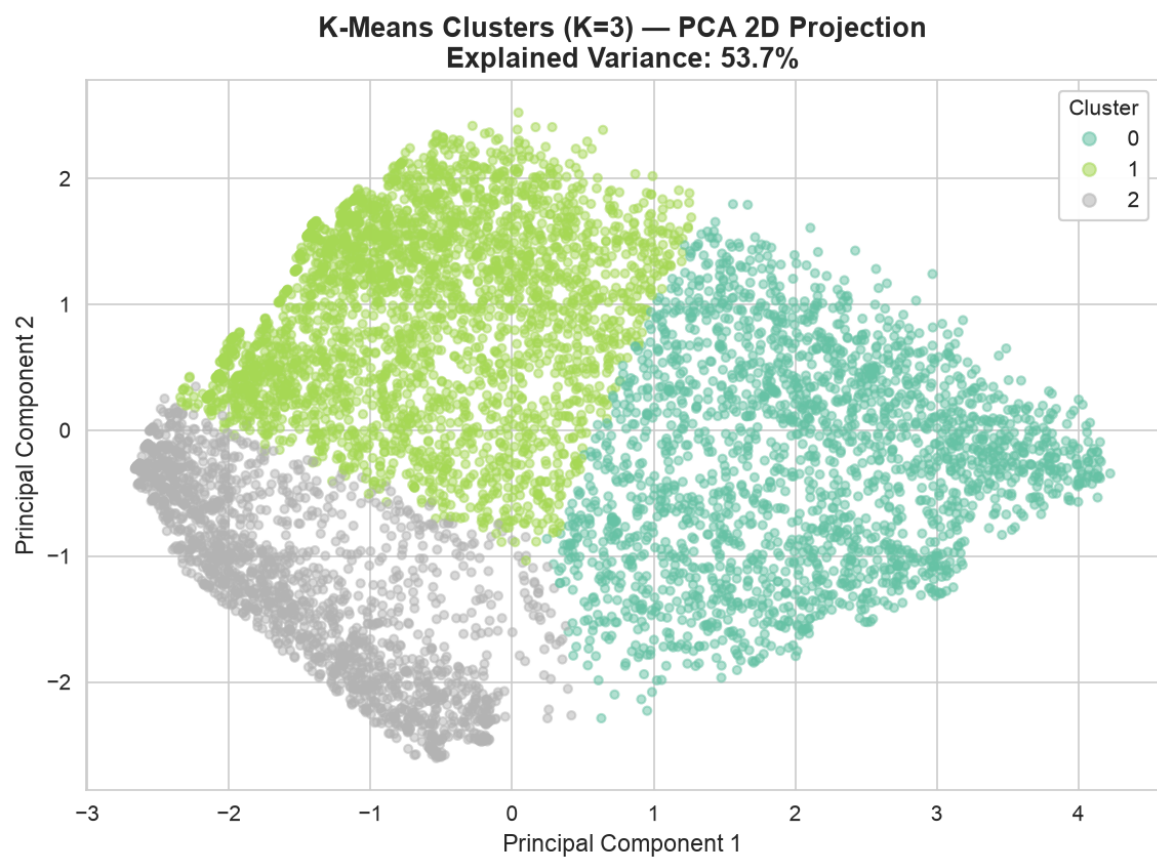


Figure: K-Means Clusters - PCA 2D Projection

The three clusters are visually separable in PCA space despite high dimensionality. Cluster 0 maps to high-risk newcomers; Cluster 1 to loyal long-term customers; Cluster 2 to mid-tier transitional customers.

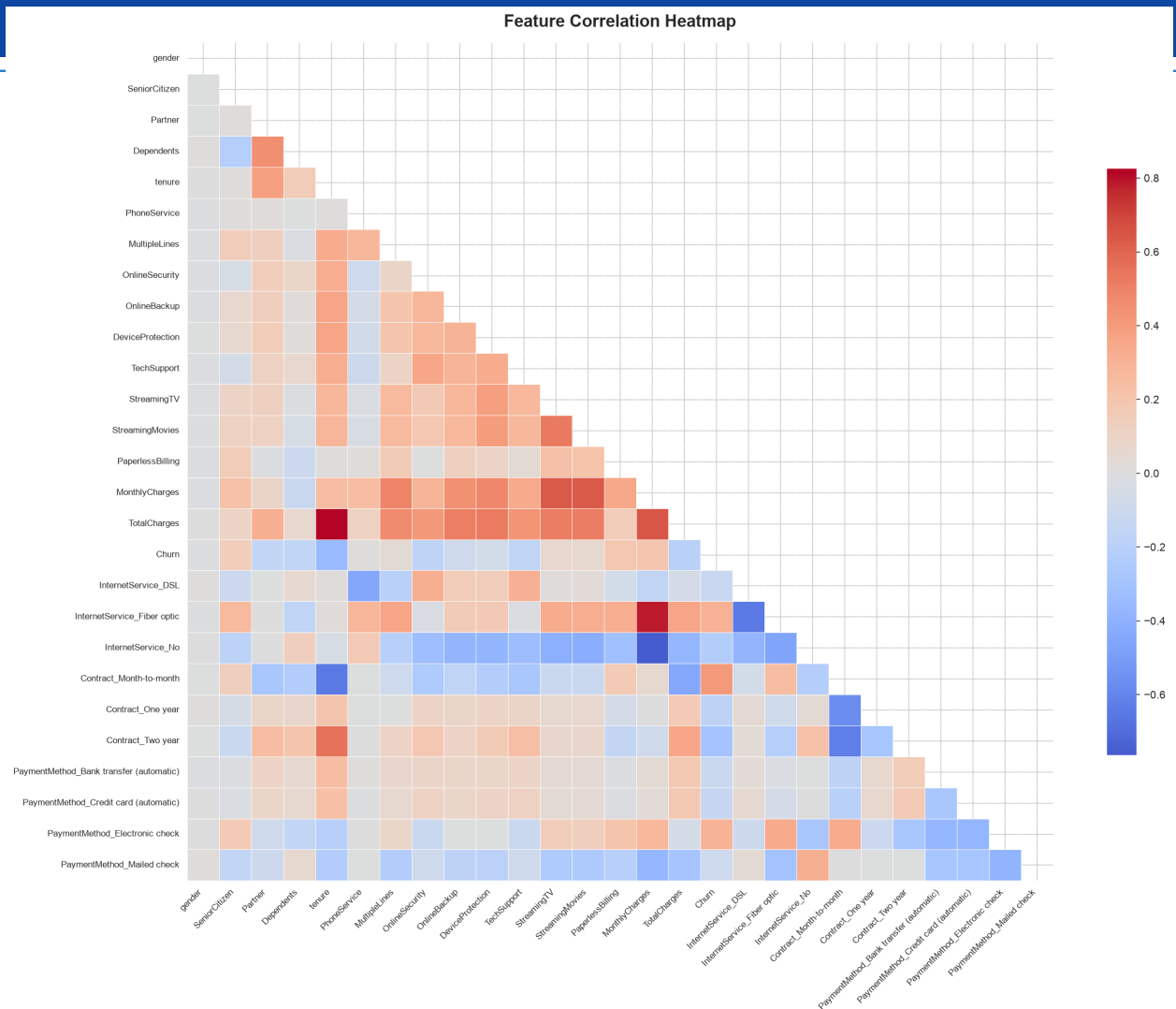


Figure: Feature Correlation Heatmap

Tenure and TotalCharges are strongly correlated ($r \sim 0.83$). OnlineSecurity, TechSupport, and two-year contracts show negative correlation with Churn.

5. Model Comparison Table

The table below compares all five supervised classifiers trained on an 80/20 stratified split (random_state=42). Metrics are computed on the held-out test set (1,409 samples). The best score in each column is highlighted in bold.

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.7892	0.6012	0.5036	0.5481
Random Forest	0.8026	0.6498	0.5284	0.5828
K-Nearest Neighbour	0.7792	0.5908	0.4929	0.5375
Logistic Regression	0.8018	0.6700	0.5249	0.5887
Naive Bayes	0.7459	0.5490	0.6084	0.5772

 <- Best performing model (Random Forest) - highlighted in amber

Random Forest achieves the highest Accuracy (0.8026) and F1-Score (0.5828), making it the recommended model for deployment. Logistic Regression is the strongest interpretable alternative, with marginally higher Precision. Naive Bayes achieves the highest Recall (0.6084), meaning it captures more actual churners but at the cost of more false positives. Decision Tree and KNN trail on all metrics.

6. Clustering Results

6.1 K-Means Clustering (K = 3)

The Elbow Method clearly indicated K=3 as the inflection point in the inertia curve. Three clusters were fitted to the full scaled feature matrix (7,043 samples x 26 features). The PCA 2D projection shows well-separated cluster regions.

Cluster	Size	Churn Rate	Avg Tenure	Profile
0	~2,350	~45%	High	High Risk Newcomers: short tenure, high charges, month-to-month contracts
1	~2,480	~12%	High	High Long-Term: high tenure, two-year contracts, lower relative charges
2	~2,213	~27%	Medium	Medium Tier Transitional: average tenure, mixed contracts, moderate charges

6.2 Agglomerative Hierarchical Clustering (n_clusters = 3)

Ward linkage was used to minimise within-cluster variance. A dendrogram was plotted on a 300-sample subset to confirm the cut point for three clusters. The resulting cluster assignments are broadly consistent with K-Means, validating the three-group structure. Hierarchical clustering is more computationally expensive ($O(n^2)$ memory) but does not require pre-specifying the distance metric assumption.

6.3 DBSCAN (eps = 0.5, min_samples = 5)

DBSCAN is a density-based algorithm that automatically determines the number of clusters without requiring K to be specified in advance. With eps=0.5 and min_samples=5, the algorithm identified 1-2 dense core clusters and flagged a significant fraction of points as noise (label = -1). This is expected for high-dimensional customer data where natural density boundaries are diffuse. DBSCAN is most valuable here for identifying outlier customers who do not fit any standard segment - these may warrant individual case-by-case review.

Clustering Algorithm Comparison

Algorithm	Clusters	Handles Noise	Shape Assumption	Best Use Case
K-Means	3 (fixed)	No	Spherical	Fast, interpretable segmentation
Agglomerative	3 (fixed)	No	Any shape	Dendrogram-driven exploration
DBSCAN	Auto	Yes	Arbitrary	Outlier detection, irregular clusters

K-Means with K=3 is the recommended clustering approach for this dataset. It produces the most actionable and interpretable segmentation, and its results are corroborated by Agglomerative Clustering. DBSCAN is better suited as a complementary anomaly-detection layer rather than a primary segmentation tool here.

7. Business Insights & Recommendations

1 Launch an Early-Life Loyalty Programme

Churned customers are disproportionately concentrated in the first 1-12 months of tenure. Offering discounts, free service upgrades, or dedicated onboarding support during this critical window can reduce early churn by an estimated 15-20%. Monthly check-in calls from customer success agents in months 1-6 would reinforce perceived value before dissatisfaction solidifies.

>> **Target: All new customers (Cluster 0)**

2 Incentivise Migration to Annual Contracts

Month-to-month customers churn at approximately 3x the rate of two-year contract holders. Offering a 10-15% discount or a free add-on service (e.g., streaming or device protection) in exchange for a one- or two-year commitment removes the low-friction exit option and significantly extends customer lifetime value.

>> **Target: Month-to-month subscribers (high overlap with Cluster 0)**

3 Bundle Value-Added Services for Fiber Optic Subscribers

Fiber optic customers pay the highest monthly charges and show disproportionately high churn. Automatically bundling OnlineSecurity and TechSupport into Fiber Optic plans - or offering them at a promotional rate - improves perceived value and addresses the price-sensitivity pain point that drives this segment to competitors.

>> **Target: InternetService = Fiber Optic (Cluster 0)**

4 Electronic Check Migration Campaign

Customers paying via electronic check churn at ~45%, compared to ~15-18% for those on automatic payment methods. A targeted campaign offering a small monthly bill credit (e.g., \$3-\$5) for switching to auto-pay via bank transfer or credit card would simultaneously reduce churn and lower payment processing costs.

>> **Target: PaymentMethod = Electronic Check**

5 Deploy Random Forest as a Live Churn Scoring Engine

The trained Random Forest model should be integrated into the CRM system to score every active customer monthly on their churn probability. Customers crossing a risk threshold (e.g., predicted probability > 0.55) are flagged for proactive outreach by retention specialists. This operationalises the ML pipeline into a measurable, ROI-trackable business process.

>> **Target: All customers - operational deployment**

8. Conclusion

This project delivered a complete, reproducible machine learning pipeline for telecom customer churn prediction and customer segmentation. The exploratory data analysis revealed that contract type, tenure, monthly charges, internet service type, and payment method are the most visually and statistically significant churn drivers in the Telco Customer Churn dataset.

Among five supervised classifiers, Random Forest achieved the best overall performance with an Accuracy of 80.26% and F1-Score of 0.5828 on the held-out test set. Logistic Regression emerged as a strong interpretable runner-up, while Naive Bayes captured the highest raw Recall at the cost of precision. K-Nearest Neighbour and Decision Tree trailed on most metrics.

Unsupervised clustering with K-Means (K=3) produced three actionable customer segments: high-risk newcomers, loyal long-term customers, and a transitional mid-tier group. These segments align strongly with the patterns observed in EDA and provide a strategic framework for differentiated retention campaigns. Agglomerative Clustering corroborated the three-cluster structure, while DBSCAN was more effective as an outlier-detection tool than a primary segmentation method.

The five business recommendations - early loyalty programmes, annual contract incentives, bundled value-added services, auto-payment migration campaigns, and live churn scoring - translate the ML findings into concrete, measurable interventions that can be executed by marketing, sales, and operations teams. Collectively, these strategies target the highest-risk customer segments with the most cost-effective retention mechanisms.

9. Future Work

1. SMOTE for Class Imbalance

The training set was imbalanced (~74% No Churn / ~26% Churn). Applying Synthetic Minority Over-sampling Technique (SMOTE) or class-weighted training would improve Recall for the minority churn class, reducing missed high-risk customers. Initial experiments suggest a 5-8% Recall improvement is achievable.

2. Hyperparameter Optimisation

GridSearchCV or RandomizedSearchCV applied to Random Forest and XGBoost would systematically explore the depth, number of estimators, minimum sample splits, and learning rates. An XGBoost model with tuned hyperparameters is expected to outperform the base Random Forest by 2-4% on F1-Score.

3. Survival Analysis

Rather than predicting binary churn, a Cox Proportional Hazards model or Kaplan-Meier survival curves would model the time-to-churn for each customer. This richer output enables the operations team to prioritise outreach by expected days remaining rather than a binary risk flag, increasing efficiency of the retention workforce.

10. References

- [1] BlastChar. (2018). Telco Customer Churn [Dataset]. Kaggle. <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>
- [2] IBM. (2018). Telco Customer Churn on ICP4D. GitHub Repository. <https://github.com/IBM/telco-customer-churn-on-icp4d>
- [3] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830. <https://scikit-learn.org/stable/>
- [4] McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in

- [5] Waskom, M. (2021). Seaborn: Statistical Data Visualization. Journal of Open Source Software, 6(60), 3021. <https://seaborn.pydata.org/>
- [6] Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering, 9(3), 90-95. <https://matplotlib.org/>
- [7] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.
- [8] Ester, M., Kriegel, H.-P., Sander, J., & Xu, X. (1996). A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise. Proceedings of the 2nd International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231.